

1078

I-125 Plaque Brachytherapy for Choroidal Melanoma: Mature Single-institution Outcomes

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Purpose/Objective(s): We report a series of patients treated with I-125 plaque brachytherapy for choroidal melanoma. The specific aims of the analysis were to evaluate local tumor control, overall and metastasis-free survival, loss of visual acuity versus loss of subjective vision, and describe the need for secondary enucleation.

Materials/Methods: We identified 317 patients with choroidal melanoma managed by I-125 plaques from 1991 to 2011. The selected tumors had been prescribed 85 Gy to the prescription point per COMS protocol. We recorded tumor and treatment characteristics, mortality, rate of local recurrence (LR), development of metastatic disease, need and reason for secondary enucleation, changes in visual acuity, and subjective vision loss (new patient-reported loss between previous time point and current time point/total patients at current time point). Kaplan-Meier survival curves were used to calculate overall and metastasis-free survival. Log-rank analysis was used to determine factors associated with overall survival.

Results: Mean tumor thickness was 3.57 mm (range, 1-12 mm) and mean maximum basal tumor diameter was 11.00 mm (range, 4-20 mm). Ciliary body invasion was seen in 28/317 patients. Using a mean plaque size of 16.91 mm (range, 4-24 mm) the mean calculated dose of 87.64 Gy was given over a mean of 144.9 hours. One and 5-year overall survival were 98.7% and 77.4%, respectively. One and 5-year metastasis-free survival were 99.7% and 93.4%, respectively. Factors associated with overall survival ($p < 0.05$) included age at treatment, maximum basal tumor diameter, and retinal detachment. Five-year LR free survival was 97.8%. There were 20 secondary enucleations total. LR was cause for secondary enucleation in 10/317 cases (all LR were enucleated); 3/10 of these LR had pre-treatment invasion into the ciliary body. Pain was the main reason for secondary enucleation in 10/317 cases. Visual acuity in the treated eye of equal or better than 20/200 was seen in 87.9%, 77.3%, 54.1%, 46.7%, and 40% of patients pre-treatment, and 1, 5, 10, and 15 years post-treatment, respectively. Subjective vision loss (within treated eye) was reported in 45.0%, 31.0%, 22.3%, 18.2%, and 16.7% of patients at pre-treatment, 1, 5, 10, and 15 years follow-up.

Conclusions: Curative intent I-125 plaque brachytherapy provides a high 5-year disease-free survival and low risk of LR. Enucleation was required in less than 6.5% of cases, with pain and local recurrence each accounting for half of the cases. Nearly one third of tumors requiring secondary enucleation due to LR had pre-treatment ciliary body invasion. Visual acuity showed a gradual decline up to 15 years post-treatment, with subjective vision loss noted predominantly in the first 5 years of follow-up.

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1079

Evaluating the Relationship Between Prescription Dose and Overall Survival Among Patients Treated With I-125 Brachytherapy for Posterior Uveal Melanoma

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Purpose/Objective(s): The Collaborative Ocular Melanoma Study (COMS) group has performed a series of randomized controlled trials to evaluate the efficacy of episcleral iodine-125 plaque brachytherapy for the

treatment of posterior uveal melanomas. In the COMS study of medium sized choroidal melanomas, 8500 cGy was prescribed to 5 mm or to the tumor apex height if greater than 5 mm. At our institution, we have typically used brachytherapy doses ranging from 6300 cGy to 8500 cGy prescribed to the tumor apex height without a 5 mm minimum prescription height. We sought to evaluate if tumor prescription dose had any effect on overall survival.

Materials/Methods: Two hundred fifty patients were identified who underwent I-125 episcleral plaque brachytherapy for choroidal and ciliary body melanoma at Duke between 1987 and 2009. Parameters retrospectively collected included tumor dimensions (basal diameter and height), baseline visual acuity, and comorbidities prior to plaque therapy. Outcomes collected include follow-up visual acuity, time to ophthalmologic morbidity, time to local failure, time to distant metastases, and time to death. Univariate and subsequent proportional hazards model (PHM) analysis was performed.

Results: Two hundred twenty-one patients from 1987-2009 had brachytherapy prescribed using iodine 125 episcleral plaque brachytherapy to the choroid and/or ciliary body and had tumor dimensions and overall survival data available. The mean, median and interquartile range of brachytherapy prescription doses were 7,669 cGy, 7,541 cGy, and 6,777 cGy-8410 cGy, respectively. Tumor size ranged from 3.4 mm-16.3 mm maximum basal diameter and 1.4 mm-10.8 mm height. Median overall survival time was 12.3 years. By the univariate analysis, survival was strongly correlated with maximum basal tumor diameter ($p = 0.001$) and age at time of procedure ($p < 0.001$). Tumor height ($p = 0.992$) and prescription dose ($p = 0.575$) were not associated with survival. Based on univariate analysis, a PHM analysis including maximum basal tumor diameter, age at time of procedure, and prescription dose, found that prescription dose was still not associated with overall survival time ($p = 0.310$). Forty-two patients in the cohort received a prescription dose of more than 8,500 cGy. Prescription dose more than 8,500 cGy was not associated with improved overall survival time in univariate ($p = 0.975$) or separate PHM analysis including maximum basal tumor diameter and age at time procedure ($p = 0.686$).

Conclusions: Treating posterior uveal melanomas with a prescription dose of less than 8,500 cGy does not alter overall survival. Further, by our protocol, tumors less than 5 mm in height may not require prescription beyond tumor apex. Reducing collateral dosing to the retina and optic nerve may reduce tissue toxicity and improve visual and anatomic outcomes in uveal melanoma.

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1080

Long-term Update on Prospective, Randomized Trial: Charged Particles Versus Iodine-125 Plaque Therapy in the Management of Uveal Melanoma

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Purpose/Objective(s): To provide long-term outcomes from the only prospective randomized dynamically balanced trial comparing helium ion (charged particle) therapy versus iodine-125 plaque therapy for the treatment of uveal melanoma.

Materials/Methods: From 1985 to 1991, 184 patients met eligibility criteria and were randomized to helium ion RT (86 patients) versus I-125 brachytherapy (98 patients). Patients were stratified by tumor diameter, thickness, distance to disc/fovea, anterior extension, and visual acuity. Tumors close to the optic disc were included. Most tumors treated were large or medium in size, and only small tumors with demonstrated growth were treated. Tumors were <15 mm in max basal diameter and <10 mm in thickness. There was an unexpected imbalance in age, with mean age for patients randomized to plaques 58 years and helium 53 years ($p = 0.02$). A minimum tumor dose of 70 GyE was delivered to the tumor apex